

Jiayu Tang

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EDUCATION

08/2024 – 07/2026 University of Nebraska-Lincoln, United States

M.Sc., Food Science and Technology (Major), Statistics (Minor) GPA: 4.0/4.0

09/2020 - 06/2024

B.Sc. (Dual Degree), Food Science and Technology University of Nebraska-Lincoln, United States

B.Eng. (Dual Degree), Food Science and Engineering Northwest A&F University, China

GPA: 3.5/4.0

PUBLICATIONS

In Preparation:

1. **Tang J**, Adinchezor C, Grimaldo C, Zhang Y, Iwuamadi I, Rasmussen C, Rose D, Majumder K, Deehan E*. Exploring the Impact of Different Dietary Pulse Market Classes on Human Gut Microbiota and Health.
2. **Tang J**#, Rasmussen C#, Deehan E*. Effect of distinct dietary pulses market classes on gut microbiome and health markers: A systematic review of intervention trials.
3. Grimaldo C, **Tang J**, Whisenhunt L, Chai W, Deehan E*. Evaluating interrelationships between body composition and immunometabolic dysfunction using bioelectrical impedance analysis: An observational study.

Published:

1. Xiao J, Li Y, Niu L, Chen R, **Tang J**, Tong Z, Xiao C*. Effect of adding fermented proso millet bran dietary fiber on micro-structural, physicochemical, and digestive properties of gluten-free proso millet-based dough and cake. *Foods*. 2023; 12(15): 2964.
2. Deng X, Niu L, Xiao J, Guo Q, Liang J, **Tang J**, Liu X, Xiao C*. Involvement of intestinal flora and miRNA into the mechanism of coarse grains improving type 2 diabetes: an overview. *Journal of the Science of Food and Agriculture*. 2023; 103(9): 4257-4267.

#: co-first authors

*: corresponding authors

RESEARCH EXPERIENCE

08/2026 - Present Department of Infectious Diseases and Immunology
Research Assistant (Full-time) University of Florida

Supervisor: Dr. Yiquan Wang

- **WP1 – Dataset construction and QC:** Testing target-conditioned human VH–VL framework compatibility using fixed CDR, antigen, and epitope controls; building a 15-medoid panel for 3 epitopes and 9 parent antibodies with framework grafts, target structures, and epitope masks.
- **WP2 – AlphaFold 3 framework swaps:** Running 270 template-free, fixed-CDR predictions with native controls; evaluating confidence, pose/CDR RMSD, epitope recovery, contacts, and VH–VL orientation.
- **WP3 – Framework-conditioned generation:** Benchmarking Germinal and RFantibody across 2 targets and 4 frameworks under a common equal-budget external evaluation protocol.

07/2024 – 07/2026 Department of Food Science and Technology
Graduate Research Assistant University of Nebraska-Lincoln

Advisor: Dr. Edward Deehan

Human Clinical Trials:

- Coordinated a 12-week randomized, crossover clinical trial investigating dry bean market-class-specific effects on gut microbiota and cardiometabolic health markers (IRB Approved; ClinicalTrials.gov Registered)
- Co-developed a NIH-style study protocol, informed consent form, and recruitment materials with the PI; authored 4 Standard Operating Procedures (SOPs) and 2 clinical trial workflows for the Deehan Lab.
- Coordinated a human biobank study aims to collect and store blood and stool samples from volunteers along with their health information (IRB Approved).
- Lead 200+ clinic visits, manage participant recruitment, scheduling, documentation, and study supplies.
- Collected and processed stool samples, blood samples, 24-hr blood pressure data, and other dietary information, and applied data formatting, cleaning, and statistical analysis with python and R.

Systematic Review:

- Applied Large Language Model (Gemini 3) to assist with preliminary data extraction and formatting, and validated model performance with human reviewers.
- Conducted a systematic review aims to identify the effects of distinct dietary pulse market classes on gut microbiome and surrogate health outcomes in humans.
- Developed a PICOS-based research proposal, conducted systematic literature searching, and applied machine learning-based tools (ASReview, TERA-Tool) for 1,800+ reference data extraction.

04/2022 - 06/2023

College of Food Science and

Engineering

Undergraduate Research Assistant

Northwest A&F

University

Advisor: Dr. Chunxia Xiao

- Conducted in vitro fermentation experiments to evaluate how three lactic acid bacteria species improve the phenolic content, antioxidant activity, and starch digestibility of *proso* millet powder.
- Performed spectrophotometric analyses to quantify the color characteristics of gluten-free cake crumb and crust made with *proso* millet bran.
- Conducted targeted literature research and summarize key findings on coarse grain consumption across 50+ articles, contributed to a review article determining the relationship between coarse grain consumption and Type 2 diabetes.

TEACHING EXPERIENCE

01/2026 – 05/2026

Department of Food Science and

Technology

Teaching Assistant

University of Nebraska-

Lincoln

- Prepared experimental materials for FDST 458-153 and led lab sessions.
- Graded assignments for FDST 458.
- Participated in the proctoring and marking of exams for 35 students.

02/2025 – 04/2025

Department of Food Science and

Technology

Guest Lecturer

University of Nebraska-

Lincoln

- Delivered 3 lectures for FDST 498 on systematic reviews, covering database search strategies, PICOS-based study design, and the application of AI/ML tools to accelerate literature screening and data extraction.

08/2023 - 05/2024
Technology
Teaching Assistant
Lincoln

Department of Food Science and
University of Nebraska-

- Graded assignments for FDST 412 and FDST 429.
- Graded assignments for NUTR 250.
- Participated in the proctoring and marking of exams for 55 students.

09/2022 - 03/2023

Foreign

Department

Teaching Assistant (Internship)
Group

Xi'An New Oriental Education & Technology

- Organized and proctored an IELTS practice exam for 71 students.
- Supported academic planning and language exam preparation for 4 students (Z Cui, C Huang, Y Yang, H Ma).
- Coordinated event planning and recruitment initiatives; recognized as "Excellent Staff of the Month."

02/2022 - 07/2022

College of Food Science and

Engineering

Exam Proctor/Grader

Northwest A&F

University

- Graded assignments and exams for Public Speaking (COMM 209).

TALKS & PRESENTATIONS

05/2026

Nebraska Food Innovation

Center

Tang J. Exploring the Impact of Different Dietary Pulse Market Classes on Human Gut Microbiota and Health. (Public Presentation)

04/2026

Nebraska Research

Days

Rasmussen C, **Tang J**, Grimaldo C, Benson A, Rose D, Mukhopadhyay I, Majumder K, Deehan E. Effect of Different Nebraska-Dry Bean Market Classes on Human Health and the Gut Microbiota: A Preliminary Analysis. (Poster)

04/2026

Nebraska Research

Days

Rasmussen C, **Tang J**, Deehan E. Dietary Pulse Market Class Specific Effects on Gut Microbiota, Metabolites, and Human Health: A systematic Review. (Poster)

03/2026

Missouri Valley Branch ASM Annual

Meeting

Tang J, Grimaldo C, Whisenhunt L, Rasmussen C, Majumder K, Rose D, Deehan E. Effect of Different Nebraska Dry Bean Market Classes on Human Health and the Gut Microbiota: A 3-Phase Randomized Crossover Clinical Trial. (Poster)

04/2025

Food Science and Technology

Symposium

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

04/2025

Nebraska Research

Days

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

04/2025

NPOD 16th Annual Spring

Retreat

Tang J, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the Impact of Different Dietary Pulses Market Classes on Host Metabolic Health and Gut Microbiome. (Poster)

AWARDS & SCHOLARSHIPS & ASSISTANTSHIPS

2026	The Chancellor's Graduate Research Grant Award	\$400
2025 - 2026	Departmental Graduate Research Assistantship	\$25500
2024	Mary and Charles C. Cooper/Emma I. Sharpless Fellowship	\$3000
2024 - 2025	Departmental Graduate Research Assistantship	\$24500
2023	National Undergraduate Life Sciences Research - Silver Prize	¥3000
2022 - 2023	Undergraduate Merit Scholarship	¥1500
2021 - 2022	Undergraduate Merit Scholarship	¥1500
2020 - 2021	Undergraduate Merit Scholarship	¥1500

MENTORSHIP

- **Mentor & Co-supervisor**, University of Nebraska-Lincoln (August 2025 – May 2026)
 - Mentored and co-supervised one undergraduate research assistant (C Rasmussen) for conducting a systematic review, clinical trial data entry, and research ethics.
- **Mentor**, University of Nebraska-Lincoln (May 2025 – May 2026)
 - Navigated a master student (L Lin) for her academic and social aspects of graduate life and connected her to relevant resources.
- **Mentor**, University of Nebraska-Lincoln (February 2025 – May 2025)
 - Mentored literature search and review for one undergraduate student (Y Yang).
- **Mentor**, Northwest A&F University (January 2023 – June 2023)
 - Mentored in lab techniques for 2 undergraduate students (Y Zhao, J Liu).

LEADERSHIP

09/2025 – 06/2026	Graduate Student
Assembly	
Graduate Representative	University of Nebraska-Lincoln
■ Attend monthly meetings, identifying and advocating on behalf of FDST graduates regarding any issues pertaining to academic affairs.	
08/2024 – 08/2025	Department of Food Science and Technology
Graduate Student Advisory Board Member	University of Nebraska-Lincoln
■ Organized the “Graduate Student Mentoring Program” to connect each new graduate student with assigned mentors. ■ Met with the Departmental Graduate Chair monthly to report on recent questions from current graduate students.	
08/2024 – 05/2025	College of Agricultural Sciences and Natural Resources
Graduate Representative	University of Nebraska-Lincoln
■ Planned and organized three professional development workshops on professorship, the academic job market, and CV/resume writing for graduate students.	
09/2020 – 06/2022	College of Food Science and Engineering
Communications Lead	Northwest A&F University

- Led a team of 13 members to develop promotional newsletters and video content weekly for the College.
- Coordinated with other Colleges' team to co-publish newsletters and video content for the university.

SERVICE

Service to the Departments, Colleges, and Universities

08/2025

Nebraska Food for Health Center
University of Nebraska-

Volunteer
Lincoln

- Hosted an outreach session at the Ice Cream Social event to introduce and advocate for the Nebraska Food for Health Center, engaging students and faculty members in discussions about its research mission and opportunities.

08/2023 – 05/2024

College of Food Science and

Engineering

Academic Advisor
University

Northwest A&F

- Advised 19 freshmen on their academic planning, research opportunities, and career development.

10/2021 – 08/2023

College of Food Science and Engineering
Northwest A&F

Academic Coordinator
University

- Facilitated communication between 59 undergraduate students and the College of Food Science and Technology to address academic and student life concerns.
- Supported “3+1” program events and activities, academic advising sessions, and handled daily student inquiries regarding to their grades and courses.

08/2022

Office of Undergraduate Studies
Northwest A&F

Volunteer
University

- Assisted with student orientation and organized outreach activities to promote student engagement.

Service to Profession

11/2024

Junior Reviewer, *Clinical Nutrition*

Service to Community

06/2022

College of Food Science and Engineering
Northwest A&F

Volunteer
University

- Participated in Undergraduate Rural Voluntary Activities for Poverty Alleviation to deliver outreach and educational programs on food safety for rural school children and families.

PROFESSIONAL DEVELOPMENT

Certifications

- Python for Data Science, AI & Development, Coursera, October 2025
- Data Science: R Basics, HarvardX, May 2025
- Alberta Food Safety Basics, University of Alberta Health Service, February 2025
- GCP for Clinical Trials with Investigational Drugs and Biologics, CITI Program, December 2024
- Biomedical Responsible Conduct of Research, CITI Program, January 2024
- Conflicts of Interest, CITI Program, December 2023

- **Biomedical Investigators and Key Personnel, CITI Program, December 2023**

Workshops & Trainings

- **Complex Biosystems Seminar Series, University of Nebraska-Lincoln. (September 2025 – August 2026)**
- **Nebraska Food for Health Center (NFHC) Seminars, Nebraska Food for Health Center. (August 2024 – August 2026)**
- **High Performance Computing for Researchers, Holland Computing Center, University of Nebraska-Lincoln. (June 6th – 26th, 2025)**
- **Exploring Your Personal Approach to Leadership, Collaboration, and Well-Being, Clifton Strengths Institute. (January 2025 – April 2025)**
- **Professional Food Science Communication, Department of Food Science and Technology. (January 2025 – May 2025)**
- **Grant Writing Seminar, University of Nebraska-Lincoln. (March 26, 2025)**
- **Food Science and Technology Seminar, Department of Food Science and Technology, University of Nebraska-Lincoln. (August 2024 – December 2024)**
- **Peer Mentorship Program, Northwest A&F University Psychological Education Association. (January 2021 – February 2021)**

PROFESSIONAL MEMBERSHIPS & AFFILIATIONS

1. ASM Branch Membership-Missouri Valley	2025 –
2026	
2. American Society for Nutrition	2024 –
2026	
3. American Society for Microbiology	2024 –
2026	
4. International Scientific Association of Probiotics and Prebiotics Students and Fellows Association (ISAPP-SFA)	2023 –
2025	

SKILLS

- **Computational Antibody Engineering:** AlphaFold 3, Germinal, RFantibody; human VH–VL framework panels, fixed-CDR framework swaps, framework grafting, target structures, and epitope masks
- **Structural Evaluation:** AF3 confidence metrics, pose RMSD, CDR RMSD, epitope recovery, contact analysis, and VH–VL orientation
- **Programming & Statistical Analysis:** Python, R, SAS; Experimental Design, Multiple Regression Analysis, Multivariate Analysis; Gemini, Claude Code, Cursor
- **Research Operations & Management:** Dataset construction and QC, clinical trials, IRB/GCP procedures, biobanking, systematic reviews; EndNote, REDCap, Notion
- **Visual Communication & Languages:** BioRender, Adobe Illustrator, Canva; Mandarin (native), English (fluent)